Outlook

RE: Benytter fugler til å plukke plast?

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Dato ma. 26.01.2026 15:14
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3 vedlegg (2 MB)

Prelikova et al 2023 Training corvids to pick up rubbish.pdf; Lamarre & Wilson 2021 Waterbird solves the string-pull test.pdf; Chaotic or lawfully evil.jpeg;

Hei

- Vi jobber med et produkt som lar fugler levere inn plast de finner i naturen, i bytte for mat. Produktet skal ved bruk av maskinlæring kun belønne fuglene som leverer inn unaturlige ting som f. eks plast eller annet søppel.

Given that you are in Elektronisk systemdesign og innovasjon, I can think of two different research questions you may wish to address. Possibly your aim is to find a creative use case for machine learning, and the point where machine learning contributes is where it distinguishes the things you want the birds to find from those you don't care about. The focus would be on whether machine learning can make that distinction.

A different research question would be investigating the contexts in which machine learning can be useful, where training birds to deliver rubbish in exchange for food is merely the example you focus on. Then you would be less interested in what machine learning can achieve (you would either know or assume that it can make the relevant distinctions), you would be interested in all the complications that arise when you deal with a real world problem, rather than a toy problem in the lab.

It is quite important which is your focus. If the first, then the birds are just what happens to deliver your data set for distinguishing rubbish from other items. Everything to do with training the birds is just stuff you have to sort out to get the data set that interests you.

If instead you are interested in the context, you have a much broader problem in animal cognition and behavioural ecology, specifically optimal foraging, and machine learning is just one aspect of your methods. This here indicates you have at least some interest in the context:

- Hvordan går man frem for å starte en slik prosess, fra et økologisk perspektiv? Kan dette være skadelig, både på kort og lang sikt?

On to practicalities:

There seem to have been several attempts to train birds to pick up rubbish, usually with crows or some other member or the corvid family. I attach a paper I found that describes the basic procedure. I have not spent time looking for more. Note that the kind of rubbish the birds were trained to pick up was small, rigid stuff: https://www.ubergizmo.com/2025/11/can-crows-be-trained-to-clean-the-streets-the-idea-refuses-to-die/

https://www.smithsonianmag.com/smart-news/french-theme-park-taught-crows-pick-trash-180969996/

Whether gulls can learn the same thing is unknown. There has been far less research into the cognition of gulls. I attach a paper. In my experience, species can differ. I have trained or tried to train four parid species (coal tits, great tits, blue tits, marsh tits), two corvid species (jackdaws and magpies) and pigeons to use a touch screen. I learned the procedure with the parids. Stick some reward on the touch screen, on top of some image. Deliver separate reward whenever the bird pecks at the food stuck on the screen and therefore the image. Gradually reduce the amount of food on the screen, to less than is delivered separately, and eventually to nothing. That is called shaping.

It worked fine with all the parids and with jackdaws in the lab. The magpies pecked at the screen so long as there was even a tiny bit of food stuck to it, but never made the transition to pecking when there was only an image on the screen. The pigeons in the lab never learned, even though all the literature I found says you don't even need to shape their responses, they are downright compulsive about pecking at signals for food. That literature also reported using operant chambers, in which the screen and the food delivery apparatus were the only thing to see. My pigeons could see each other, and I wondered whether that was too much distraction. But later I heard that Nick Mackintosh had gotten free-living urban pigeons to work on touch screens. Species and the details of the procedure matter, and you may spend a lot of time failing to get the gulls to do what you want.

Assuming you can train the gulls, your biggest problem is likely to be the one Ragnhild mentioned, and related:

- Vil fuglene også begynne å ta 'søppel' som folk har stående i hager eller bærer på (tenk på de tidvis aggressive måkene på torget sommerstid)?

I very much expect they will. For the birds, this is a problem in optimal foraging. Where can they get plastic that takes the least effort per reward?

At sea, plastic is thinly distributed and therefore, on average, far from where you reward for delivery. Transporting it over those distances is costly. If a bird is at sea already, it may pay better to look for sea life.

In town, though, there is a rich source of plastic which urban gulls already raid for food: rubbish bins. I would expect gulls to raid rubbish bins for plastic, and only look elsewhere once all the rubbish bins are empty. The elsewhere may be plastic people carry. Or plastic in shops. It is not difficult to find videos of gulls walking into shops and nicking snacks. Training gulls to associate plastic with food may give more of them the idea to get both at the source, and then to deliver the empty packaging for more food.

For logistic reasons, I expect you would set up your exchange station in town. You would need to know whether urban gulls even forage at sea.

If gulls that visit your exchange station forage at sea, the most common plastic would be bags and bottles. Bottles and bags may be filled with liquid. Draining the liquid would be a prey handling skill that the birds would have to learn.

If you can train gulls to pick up rubbish from the water, they may just pick up anything that floats. One problem in picking up floating plastic is to avoid picking up any of the ecological community of sea life that floats on the surface, collectively called nekton. To be fair, I never noticed any on Trondheimsfjord, but the great Pacific garbage patch is also prime habitat for nekton, so this is a problem for proposed mechanical collectors that would sweep up anything that floats.

How does your target species transport food? Some species in their feet (raptors, so not likely relevant to you), some in their beaks, some in throat pouches. If the last, are they more likely to swallow? I think gulls favour throat pouches, but you would have to check.

Will you do this during breeding season? The longer the distance between foraging grounds and nest, the more likely it is that birds will collect multiple items before returning. If some items are food and some rubbish, and you train them to pick up rubbish, will the birds sort items before feeding their young? If they try to go to the exchange station first, and they need to put items down to sort them, other birds will at least try to take whatever is edible. That would discourage birds from going to your exchange station first. Will they remember to sort rubbish from food before feeding their young? As far as I know, normal procedure is to regurgitate food directly into the mouths of the young, rather than to let the young pick it up.

Do you have a way to deal with birds just hanging out at the exchange station, waiting for others to deliver rubbish, and then picking up the rewards?

How does the exchange station collect items that are not small and rigid? If it accepts bottles, what prevents birds from going in after the bottles to deliver them repeatedly? If bags, is there an opening large enough for bags to fall in? Is there something that pulls them in? If so, how do you prevent that mechanism from pulling in birds?

What you propose would be an animal experiment, on free-living animals. You would need to get permission from Direktoratet for Naturforvaltning. If this is your Bachelor thesis, to be delivered this May, I don’t think you have enough time. This bureaucracy is not fast. And you may need a collaborator who has experience with this kind of work.

If your focus is not so much on the capabilities of machine learning, but on the complications that arise when applying it to the real world, then you have identified an application with plenty of complications, enough that this problem may well be beyond the scope of a Bachelor thesis, unless you make this theoretical and dive into the relevant literature. If system design includes being mindful of consequences, then much of your task is to find people who can point you in the right direction. I would recommend checking for behavioural ecologists at NTNU, especially someone working in optimal foraging, or the foraging of urban birds.

Regards

Robert Biegler